

1. Motivation

Transcribing historical handwritten documents at scale is a persistent challenge in digital humanities and intelligent document processing. The personal diaries of **Julia Newberry**^[1] (1869–1871), held in the Newberry Library’s online collection, exemplify this difficulty: dense, flowing cursive script with irregular line spacing and overlapping ascenders and descenders. Before any machine learning model can learn to read handwriting, the raw manuscript page must be decomposed into individual, clean text lines. This data preparation step directly determines whether a transcription model can train effectively.

Core Challenge

Full-page transcription models demand large memory, extensive annotated data, and powerful hardware. A smarter decomposition strategy was essential to make this project feasible on local hardware.

2. Dataset

The dataset consists of scanned pages from Julia Newberry’s personal diary, written in flowing cursive—a style challenging even for trained human readers unfamiliar with 19th-century handwriting conventions.

Statistic	Value
Total manuscript pages	171
Writing style	Flowing cursive
Date range	1869–1871
Line images extracted	3,701
Avg. lines per page	22
Processing time (244 pp.)	less than 4 mins

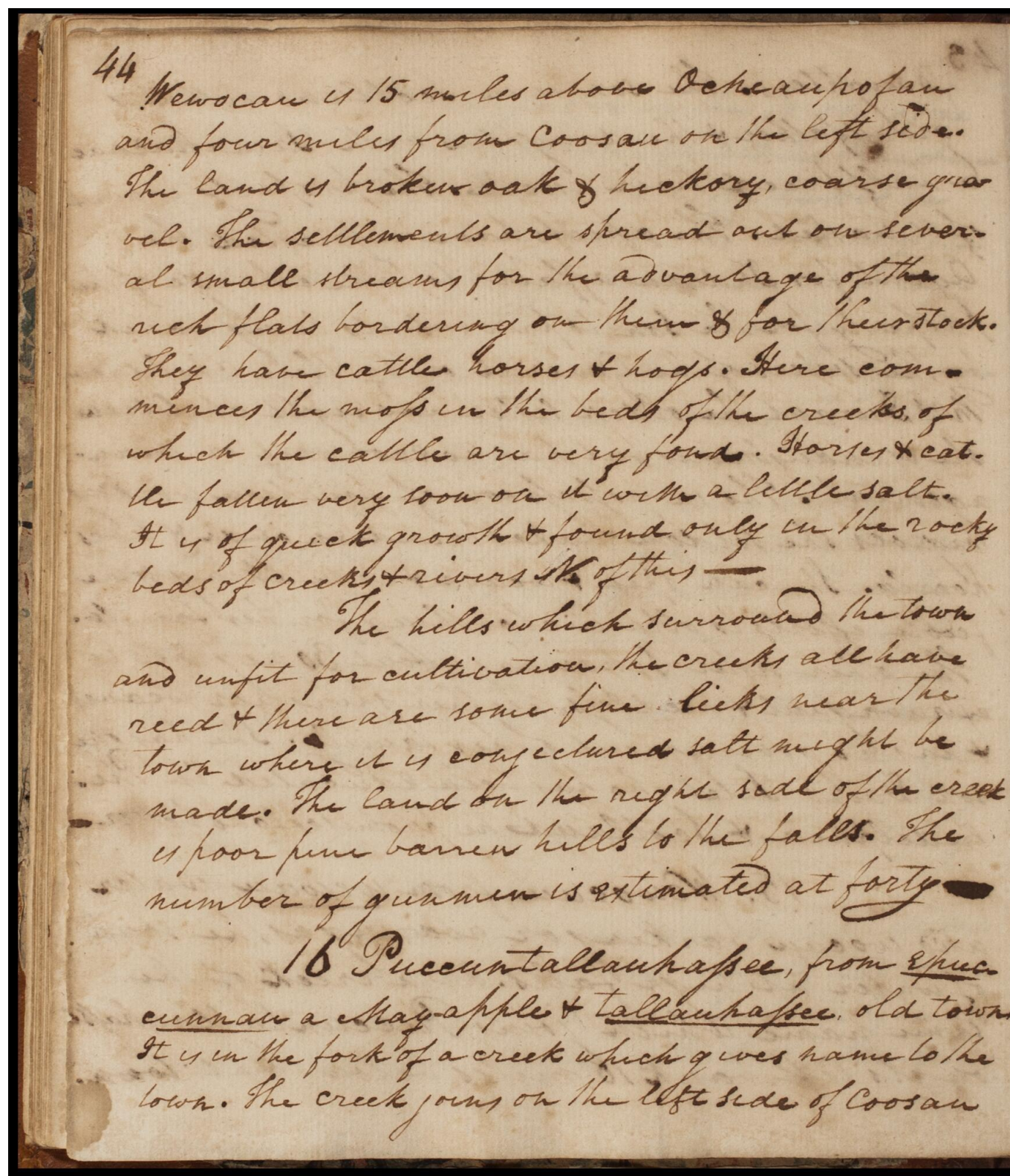


Figure 1. Sample Image of Julia Newberry Data Set

Dataset Credit

Sourced and curated by **Micah Hesketh** from the Newberry Library online collection (freely available, no copyright restrictions).

3. Why Line Segmentation?

Decomposing each page into individual text lines offers concrete advantages over full-page transcription:

Factor	Full-Page	Line-Level
Image dimension	~ 1705 × 2000	~ 517 × 49
Memory use	High	Reduced
Layout learning	Required	Not needed
Training data	Very large	Manageable
Local hardware	Infeasible	Feasible

4. Methods Compared

- Supervised: dhSegment-torch^[2] Trained on 13 manually annotated pages. The model **overfitted severely** to the small training set and failed to generalize to unseen manuscript pages. Scaling up annotation was not feasible within the project timeline.
- Unsupervised: Seam Carving**^[3] (**Selected**): An energy minimization technique that finds the lowest-cost horizontal path through whitespace between text lines. Requires **no annotations**, handles cursive overlap naturally, and is computationally efficient.

Why not connected components? Cursive characters from adjacent lines frequently overlap (e.g., descender of ‘g’ touching the ascender of ‘h’ below). Connected component analysis merged these into single regions. Seam carving navigated around overlaps by following background regions.

5. The Seam Carving Energy Function

The seam carving algorithm assigns a cost to each pixel:

$$E(y, x) = 1500 I_{\text{bin}}(y, x) + \left| \frac{\partial I_{\text{bin}}}{\partial y} \right| + \left| \frac{\partial I_{\text{bin}}}{\partial x} \right| + P_{\text{edge}}(y, x)$$

Term	Purpose
$1500 \cdot I_{\text{bin}}$	Strongly penalizes ink pixels — seams avoid text
$\left \frac{\partial I_{\text{bin}}}{\partial y} \right , \left \frac{\partial I_{\text{bin}}}{\partial x} \right $	Penalty near ink edges — keeps seams in clear background
P_{edge}	Penalizes left/right margins — prevents seams escaping into dark borders

Dynamic programming recurrence:

$$M(y, x) = E(y, x) + \min \begin{cases} M(y-1, x-1) + \lambda, \\ M(y, x-1), \\ M(y+1, x-1) + \lambda \end{cases}$$

$\lambda = 1.0$ movement penalty applied when the seam shifts up or down between columns.

Text Annotation (Ink-Density Heuristic)

Each line image is paired with its transcription text using a proportional ink-density ratio:

$$t_i = \left\lfloor \frac{w_i}{\sum_{j=1}^n w_j} \times T \right\rfloor \quad w_i = \text{ink amount in line } i, \quad T = \text{total characters on page}$$

6. Pipeline: Page → Line Segments

Step 1: Border Cropping Raw scanned pages contain rough dark edges from the scanning process.

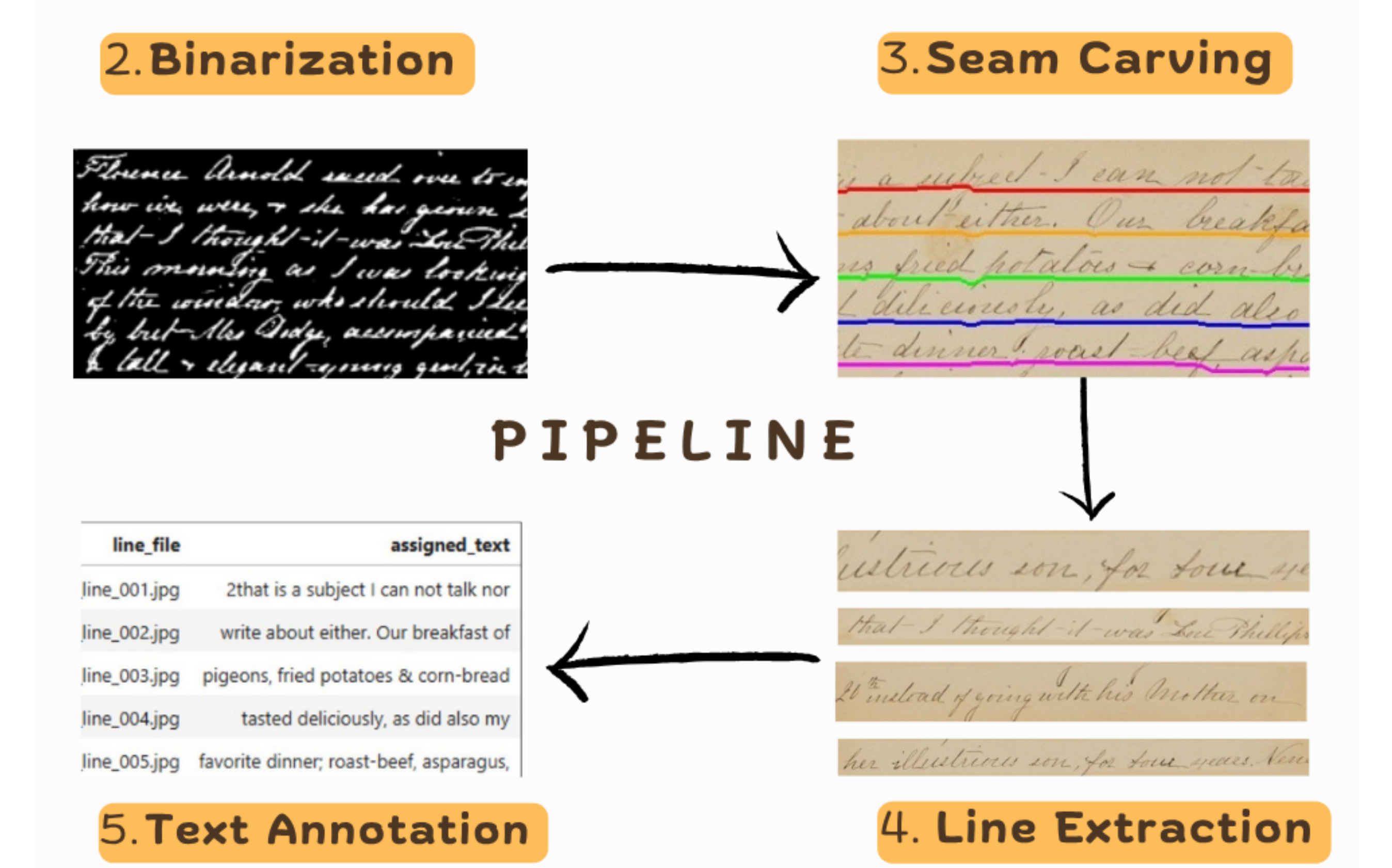


Figure 2. Overview of pipeline: from binarized page scans to annotated line segment images.

7. Results

Unsupervised seam carving is a **practical, annotation-free, and computationally efficient** approach to line segmentation for historical cursive manuscripts. The resulting 3,701 annotated line images provide a clean, structured foundation for downstream transcription model training.

Metric	dhSegment	Seam Carving
Generalization	Poor	Strong
Cursive overlap	Moderate	Excellent
Annotation needed	manual	None
Speed	Slow	<4 min / 244 pp.
Line images out	—	3,701

8. Conclusion

- Limitations:** Segmentation quality degrades on very dense, heavily overlapping pages. Text-to-line annotation relies on a heuristic, not ground-truth alignment.
- Future directions:** Tune energy parameters for dense handwriting; explore learning-based alignment methods; develop ground-truth line annotations for quantitative evaluation.
- Key Takeaway:** Thoughtful data preparation is not a preliminary step—it is a **core research contribution** that determines the ceiling of any downstream model.

9. Acknowledgments

The authors thank the Newberry Library for providing access to the Julia Newberry diary collection.

[1] N. L. Digital Initiatives, “About this data,” Apr. 2026.
[2] S. Ares Oliveira, B. Seguin, and F. Kaplan, “dhsegment: A generic deep-learning approach for document segmentation,” in *Frontiers in Handwriting Recognition (ICFHR)*, 2018 16th International Conference on, pp. 7–12, IEEE, 2018.
[3] R. Saabnii, A. Asi, and J. El-Sana, “Text line extraction for historical document images,” *Pattern Recognition Letters*, vol. 35, pp. 23–33, Jan. 2014.